

Lecture 1

The Poincaré group and the classification of particles

Objectives. *The finale executes the program. Adding translations to the Lorentz group gives the Poincaré group — a semidirect product in the sense of Lecture 2, realised as 5×5 matrices — and the fields-versus-states principle of Lecture 14 says that its infinite-dimensional unitary irreducible representations are the elementary particles. We classify them by Wigner’s method: diagonalise the translations, watch momenta sweep out orbits — mass shells and the light cone, with their invariant measures derived separately — and reduce everything to the little group of one reference momentum: $SU(2)$ for massive particles, whence spin; the double cover of the Euclidean plane group for massless ones, whence helicity and the photon’s two polarisations, alongside the mathematically allowed continuous-spin branch. For massive particles the Casimirs recover mass and spin; for finite-helicity massless particles both Casimirs vanish and helicity remains a little-group character. A one-page retrospect closes the course by stating its boundary: the representation-theoretic vocabulary needed for $SU(3)$ has been prepared, while particle-physics $SU(3)$ itself is assigned to explicit further reading.*

1.1 The Poincaré group

The interval is not only rotation- and boost-invariant; it is translation-invariant, since it depends on differences of events. The full symmetry group of special relativity is therefore the affine maps

$$x \mapsto \Lambda x + a, \quad \Lambda \in SO^+(1, 3), \quad a \in \mathbb{R}^4,$$

composing as

$$(a, \Lambda) (a', \Lambda') = (a + \Lambda a', \Lambda \Lambda') : \quad (1.1)$$

the *restricted Poincaré group* $\mathcal{P}^+ = \mathbb{R}^4 \rtimes SO^+(1, 3)$, Lecture 2’s semidirect product with continuous ingredients — the translations a normal subgroup, the Lorentz group acting on them by its defining representation. It is a matrix group by the affine trick: let (a, Λ) act on $(x, 1)^T \in \mathbb{R}^5$ as

$$g(a, \Lambda) = \begin{pmatrix} \Lambda & a \\ 0 & 1 \end{pmatrix},$$

a closed subgroup of $GL(5, \mathbb{R})$; block multiplication reproduces (1.1). Differentiating, the Lie algebra consists of $\begin{pmatrix} W & b \\ 0 & 0 \end{pmatrix}$ with $W \in \mathfrak{so}(1, 3)$, $b \in \mathbb{R}^4$: ten dimensions, with four new abstract generators p_μ (translation along e_μ) joining L_a and K_a . The new brackets are two block multiplications:

$$[p_\mu, p_\nu] = 0, \quad [W, p_b] = p_{Wb} \quad \text{for } W \in \mathfrak{so}(1, 3), \quad (1.2)$$

the second saying that the four translations transform as a four-vector; spelled out on the basis,

$$[L_a, p_0] = 0, \quad [L_a, p_b] = \epsilon_{abc} p_c, \quad [K_a, p_0] = p_a, \quad [K_a, p_b] = \delta_{ab} p_0.$$

The physics enters through Lecture 9's dictionary, now once and for all. For an honest strongly continuous unitary representation U with derived representation π , define the self-adjoint translation observables

$$\widehat{P}_\mu := i\pi(p_\mu), \quad U(te_\mu) = e^{t\pi(p_\mu)} = e^{-it\widehat{P}_\mu}. \quad (1.3)$$

Thus p_μ is an abstract Lie-algebra element, $\pi(p_\mu)$ is its generally unbounded skew-adjoint derived operator, and $\widehat{P}_\mu$ is the self-adjoint physical energy-momentum observable. With $\widehat{J}_a = i\pi(L_a)$ and $\widehat{K}_a = i\pi(K_a)$, the abstract brackets translate on a common invariant domain to

$$[\widehat{P}_\mu, \widehat{P}_\nu] = 0, \quad [\widehat{J}_a, \widehat{P}_0] = 0, \quad [\widehat{J}_a, \widehat{P}_b] = i\epsilon_{abc} \widehat{P}_c, \quad [\widehat{K}_a, \widehat{P}_0] = i\widehat{P}_a, \quad [\widehat{K}_a, \widehat{P}_b] = i\delta_{ab} \widehat{P}_0.$$

Equivalently set $\widehat{M}_{0a} = \widehat{K}_a$ and $\widehat{M}_{ab} = -\epsilon_{abc} \widehat{J}_c$. With $\eta = \text{diag}(+1, -1, -1, -1)$, the covariant observable brackets used below are

$$\begin{aligned} [\widehat{M}_{\alpha\beta}, \widehat{P}_\nu] &= i(\eta_{\alpha\nu} \widehat{P}_\beta - \eta_{\beta\nu} \widehat{P}_\alpha), \\ [\widehat{M}_{\alpha\beta}, \widehat{M}_{\rho\sigma}] &= i(\eta_{\alpha\rho} \widehat{M}_{\beta\sigma} - \eta_{\alpha\sigma} \widehat{M}_{\beta\rho} - \eta_{\beta\rho} \widehat{M}_{\alpha\sigma} + \eta_{\beta\sigma} \widehat{M}_{\alpha\rho}). \end{aligned} \quad (1.4)$$

Stone's theorem supplies the self-adjoint closures. In this notation the four-momentum is a four-vector without collapsing the three levels of the dictionary.

One paragraph of thesis before the work. A relativistic quantum system carries, by Lecture 11's reasoning, a *projective* unitary representation of $\mathcal{P}^+$. The cure is the one rehearsed twice already: pass to the universal cover. Replacing $\text{SO}^+(1, 3)$ by $\text{SL}(2, \mathbb{C})$ (acting on translations through Φ) gives

$$\mathcal{P} := \mathbb{R}^4 \rtimes \text{SL}(2, \mathbb{C}),$$

topologically $\mathbb{R}^4 \times S^3 \times \mathbb{R}^3$ by Lecture 14 — simply connected — and covering $\mathcal{P}^+$ two-to-one. We now use the Poincaré-specific cohomological lifting result (import I10): for the connected group $\mathcal{P}^+$, the Poincaré Lie algebra satisfies $H^2(\mathfrak{p}, \mathbb{R}) = 0$, and every continuous projective unitary representation pulls back to an honest strongly continuous unitary representation of the simply connected cover $\mathcal{P}$, inducing the same action on rays. This is not a consequence of semisimplicity — $\mathfrak{p}$ is not semisimple — and is distinct from Lecture 14's Lorentz-factor lifting. Thus, unlike the Galilei case with its irremovable mass phase, no central charge survives here. Wigner's definition closes the circle: an *elementary particle* is a system whose state space carries an irreducible such representation — elementary because irreducible: no invariant subspace, no way to take it apart. Lecture 14 proved these representations must be infinite-dimensional; Lecture 13 taught us not to flinch. The rest of the lecture classifies them.

Scope signal. The ray action is projective; import I10 replaces it by an honest strongly continuous unitary representation of the cover on a separable Hilbert space. Its derived operators are unbounded and share only stated invariant domains. This is a different category from Lecture 14's finite-dimensional nonunitary field representations, and finite character, trace, and dimension theorems do not transfer automatically.

1.2 Momentum and the orbit picture

Wigner's method begins where the structure suggests: with the abelian normal subgroup. The irreducible unitary representations of $\mathbb{R}^4$ are its characters $a \mapsto e^{-i\eta(p,a)}$, labelled by points p of momentum space — the plane waves: Lecture 12's "Fourier series is character theory" for the circle, four noncompact times over. Diagonalising the commuting unitaries $U(a)$ simultaneously is the imported SNAG theorem (Stone–Naimark–Ambrose–Godement): a strongly continuous unitary representation of the locally compact abelian group $\mathbb{R}^4$ on a separable Hilbert space has a unique projection-valued measure on its dual such that $U(a) = \int e^{-i\eta(p,a)} dE(p)$. This is a spectral integral, not a finite-dimensional simultaneous eigenbasis (Berndt, Theorem 5.2, p. 71; Barut–Rączka, Ch. 5). It therefore organises states by momentum, and the Lorentz half of the group shuffles the labels: from (1.1),

$$U(A) U(a) U(A)^{-1} = U(\Lambda a), \quad \Lambda = \Phi(A),$$

so if a state has momentum p then $U(A)$ carries it to one of momentum Λp — the computation is one line with $\eta(p, \Lambda^{-1}a) = \eta(\Lambda p, a)$. In an *irreducible* representation the momenta in play must therefore form a single $SO^+(1, 3)$ -orbit (heuristically clear; honestly, part of the imported theorem below). The orbits can be listed completely.

Proposition 1.1 (The orbit catalogue). *The orbits of $SO^+(1, 3)$ on $\mathbb{R}^4$ are: the mass shells $X_m^\pm = \{\eta(p, p) = m^2, \pm p^0 > 0\}$ for each $m > 0$; the two halves of the light cone $X_0^\pm = \{\eta(p, p) = 0, p \neq 0, \pm p^0 > 0\}$; the origin $\{0\}$; and the one-sheeted hyperboloids $\{\eta(p, p) = -\mu^2\}$, $\mu > 0$.*

Figure 1.1 draws the catalogue and, looking ahead to the next section, the little group each orbit carries: the forward mass shell with its $SU(2)$, the forward cone with $\widetilde{E}(2)$, the excluded spacelike sheet, and the vacuum at the origin.

Proof. The quantity $\eta(p, p)$ is invariant, and for $\eta(p, p) \geq 0$, $p \neq 0$, so is $\text{sgn } p^0$: there $|\vec{p}| \leq p^0$ or $\leq -p^0$, and the same Cauchy–Schwarz estimate that separated the components in Lecture 14 gives $|\sum_i \Lambda^0_i p^i| < \Lambda^0_0 |p^0|$, so $(\Lambda p)^0$ keeps the sign of p^0 — orthochronicity protecting causality. Conversely each listed set is one orbit: a timelike p with $p^0 > 0$ is boosted to rest, $(m, \vec{0})$, by the pure boost along $\vec{p}$ with $\tanh \chi = |\vec{p}|/p^0$; a null p with $p^0 > 0$ is rotated to point along e_3 and then boosted along e_3 — which by Lecture 14's boost algebra scales $(E, 0, 0, E) \mapsto (e^\chi E, 0, 0, e^\chi E)$ — onto any reference point of the upper cone; spacelike vectors are handled the same way. ■

Physics has named everything in the catalogue: X_m^+ is the *mass shell* of a particle of mass m , and energy $p^0 = E_p = \sqrt{m^2 + |\vec{p}|^2} \geq m > 0$ being bounded below is exactly the choice of the upper sheet (the lower sheet is reinterpreted as antiparticles — a story that belongs to field theory); the origin will be the vacuum; the spacelike orbits wait, ominously, for the end of the lecture. To do analysis on an orbit we need its invariant measure, and the shell trick of Lecture 12 delivers it one signature up.

Proposition 1.2 (The invariant measure on the mass shell). *For continuous, compactly supported f define*

$$\int_{X_m^+} f d\mu := \lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \int_{m^2 \leq \eta(p,p) \leq m^2 + \varepsilon} f(p) d^4 p = \int_{\mathbb{R}^3} f(E_p, \vec{p}) \frac{d^3 p}{2E_p}. \quad (1.5)$$

This measure is $SO^+(1, 3)$ -invariant: $\int f(\Lambda^{-1}p) d\mu(p) = \int f d\mu$.

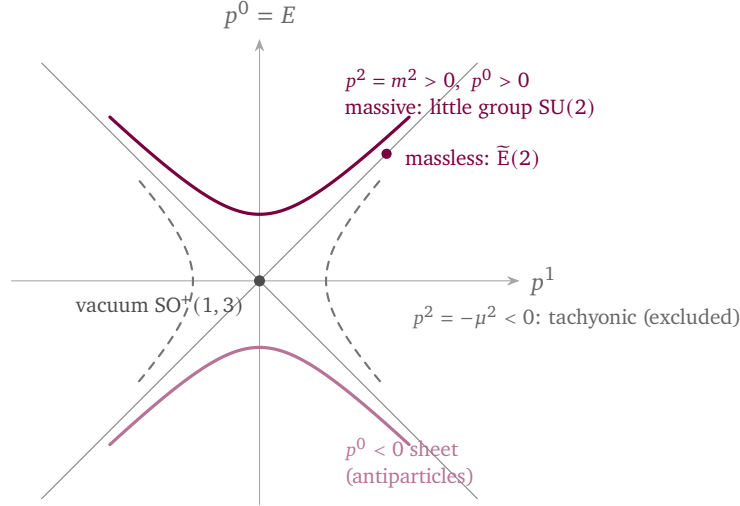


Figure 1.1: The orbits of the Lorentz group on momentum space, and the Wigner classification they carry. Lorentz transformations preserve $p^2 = (p^0)^2 - |\vec{p}|^2$ and the sign of p^0 , so they move momenta only along the level sets drawn here. The physical orbits are the forward mass hyperboloid $p^2 = m^2 > 0$, $p^0 > 0$, whose little group (the stabiliser of a rest momentum) is $SU(2)$, the cover of the geometric rotation group $SO(3)$ — its irreducible representations are the spins; and the forward light cone $p^2 = 0$, whose little group is the double cover $\tilde{E}(2)$ of the Euclidean group and whose finite-helicity representations are labelled by helicity, with a separate continuous-spin branch. The spacelike orbit $p^2 < 0$ (tachyonic) is excluded, and the origin $p = 0$ is fixed by all of $SO^+(1, 3)$ — the vacuum. A relativistic particle is an orbit together with an irreducible representation of its little group.

Proof. Invariance is manifest in the first expression: d^4p is preserved ($|\det \Lambda| = 1$) and so is the thickened shell. For the second, integrate over p^0 at fixed $\vec{p}$: the shell condition $m^2 \leq (p^0)^2 - |\vec{p}|^2 \leq m^2 + \varepsilon$ cuts out an interval of length $\varepsilon/2E_p + O(\varepsilon^2)$ around $p^0 = E_p$, and the limit collapses onto it. ■

The mysterious $2E_p$ under every integral of relativistic quantum theory is therefore not a convention but a Jacobian: the price of an invariant measure, paid once and for all.

The cone requires its own endpoint check; the preceding proposition assumed $m > 0$.

Proposition 1.3 (The invariant measure on the forward light cone). *On the punctured forward cone $X_0^+ = \{p \neq 0 : p^2 = 0, p^0 > 0\}$, the invariant measure is*

$$d\mu_0(p) = \frac{d^3p}{2|\vec{p}|}, \quad \int_{X_0^+} f d\mu_0 = \int_{\mathbb{R}^3 \setminus \{0\}} f(|\vec{p}|, \vec{p}) \frac{d^3p}{2|\vec{p}|}. \quad (1.6)$$

Proof. Start from the Lorentz-invariant distribution $d^4p \theta(p^0) \delta(p^2)$. For fixed $\vec{p} \neq 0$ its positive-energy root is $p^0 = |\vec{p}|$, and

$$\left| \frac{\partial((p^0)^2 - |\vec{p}|^2)}{\partial p^0} \right|_{p^0=|\vec{p}|} = 2|\vec{p}|.$$

Thus integrating first in p^0 gives exactly (1.6), including the factor 2. The vertex is omitted because the simple-root formula degenerates there; it is a null set, and the density is locally

integrable since in spherical coordinates $d^3p/(2|\vec{p}|) = \frac{1}{2}r dr d\Omega$. Finally d^4p , $\delta(p^2)$, and the positive sheet are preserved by $SO^+(1, 3)$, proving invariance. This is the massless norm measure used below. ■

Sheet 15, Exercise 4 checks both (1.5) and (1.6) directly under a boost in the z direction.

1.3 Little groups

The orbit tells us where the momenta live; the remaining freedom sits over a single point of it.

Definition 1.4 (Little group). Fix a reference momentum $\hat{p}$ on an orbit. The *little group* is its stabiliser in the cover, $G_{\hat{p}} = \{A \in SL(2, \mathbb{C}) : \Phi(A)\hat{p} = \hat{p}\}$.

Two computations, both two lines in Lecture 14's Hermitian model $X(p) = p^0\mathbb{K} + \vec{p} \cdot \vec{\sigma}$.

Proposition 1.5 (The massive little group). For $\hat{p} = (m, 0, 0, 0)$, $m > 0$: $X(\hat{p}) = m\mathbb{K}$, and $A(m\mathbb{K})A^\dagger = m\mathbb{K}$ says exactly $AA^\dagger = \mathbb{K}$. So $G_{\hat{p}} = SU(2)$.

The little group of a massive particle is the rotation group of its rest frame — doubly covered, as quantum mechanics insists. Lecture 10's classification is about to be promoted from theorem to definition.

Proposition 1.6 (The massless little group). For $\hat{p} = (1, 0, 0, 1)$: $X(\hat{p}) = \mathbb{K} + \sigma_3 = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$, and $AX(\hat{p})A^\dagger = X(\hat{p})$ forces the first column of A to be $(\alpha, 0)$ with $|\alpha| = 1$; the determinant then fills in the rest:

$$G_{\hat{p}} = \left\{ \begin{pmatrix} \alpha & \beta \\ 0 & \bar{\alpha} \end{pmatrix} : |\alpha| = 1, \beta \in \mathbb{C} \right\} \cong \mathbb{C} \rtimes U(1),$$

with $\text{diag}(\alpha, \bar{\alpha})$ acting on the normal subgroup $\{\beta\}$ by $\beta \mapsto \alpha^2\beta$. Since $\Phi(\text{diag}(\alpha, \bar{\alpha}))$ is the rotation by the angle of α^2 about the ray's direction, this is the double cover $\widetilde{E}(2)$ of the Euclidean group of the plane: rotations and translations of an abstract $\mathbb{R}^2$, with α and $-\alpha$ covering the same rotation.

Proof. Writing $X(\hat{p}) = 2e_1e_1^\dagger$, the condition says $(Ae_1)(Ae_1)^\dagger = e_1e_1^\dagger$, i.e. $Ae_1 = \alpha e_1$ with $|\alpha| = 1$; then $\det A = 1$ forces the diagonal $(\alpha, \bar{\alpha})$ with a free upper entry β . The semidirect structure is the matrix product $\text{diag}(\alpha, \bar{\alpha}) \begin{pmatrix} 1 & \beta \\ 0 & 1 \end{pmatrix} \text{diag}(\bar{\alpha}, \alpha) = \begin{pmatrix} 1 & \alpha^2\beta \\ 0 & 1 \end{pmatrix}$. ■

The symmetry group of a flat plane, stabilising a ray of light: an unexpected guest, whose translations will pose a physical riddle in Section 1.5. The general machine that assembles a representation of $\mathcal{P}$ from an orbit and a little-group representation is the course's final import.

Theorem 1.7 (Mackey's regular semidirect-product theorem; imported). Let $N \rtimes H$ be a second-countable locally compact group with N abelian and with regular H -action on $\widehat{N}$. Then every irreducible strongly continuous unitary representation on a separable Hilbert space is induced from a character in one H -orbit and an irreducible unitary representation of its stabiliser; the induced representation is irreducible, these exhaust the irreducibles, and equivalence is precisely conjugacy of the orbit–stabiliser data. For $\mathcal{P} = \mathbb{R}^4 \rtimes SL(2, \mathbb{C})$ these hypotheses hold: the translation subgroup is abelian, both factors are second-countable locally compact Lie groups, and the momentum action has the regular, locally closed orbits of Proposition 1.1.

Consequently an irreducible representation is determined, up to unitary equivalence, by a pair: an orbit O of Proposition 1.1 and an irreducible unitary representation σ of the little group of a reference point $\hat{p} \in O$. The representation is realised on $L^2(O, d\mu; V_\sigma)$ as a measurable induced representation.

Proof boundary and source. SNAG above supplies only the translation projection-valued measure. The orbit reduction, measurable induction, induced irreducibility, exhaustivity, and equivalence clauses are the imported theorem: Folland, *A Course in Abstract Harmonic Analysis*, 2nd ed., Theorem 6.42 and the discussion in §6.6, pp. 199–201. Computing the unitary dual of a particular little group is a further, separate task.

We import the bookkeeping — that the list is exhaustive — and build the entries that matter by hand. The catalogue is now a table: massive orbits carry $SU(2)$, so their representations are labelled (m, j) with $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$ — *mass and spin*; massless orbits carry $\tilde{E}(2)$ — *finite helicity* and continuous-spin branches, as we shall see; the origin's little group is all of $SL(2, \mathbb{C})$, whose only finite-dimensional unitary representation is trivial (Lecture 14's no-go): the trivial pair is the *vacuum*, the unique Poincaré-invariant state, and nature declines the infinite-dimensional alternatives.

1.4 Massive particles: mass and spin

Fix $m > 0$, the orbit X_m^+ with $\hat{p} = (m, 0, 0, 0)$, and a spin $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$ with Lecture 10's representation (Π_j, V_j) of the little group $SU(2)$. The construction needs one choice: a *standard boost* carrying $\hat{p}$ to each $p \in X_m^+$, and Lecture 14's square-root machinery supplies a canonical one. The matrix $X(p)/m$ is positive definite on the upper shell (its eigenvalues are $(p^0 \pm |\vec{p}|)/m > 0$) with determinant one, so

$$S_p := (X(p)/m)^{1/2} \in SL(2, \mathbb{C}) \quad \text{satisfies} \quad S_p X(\hat{p}) S_p^\dagger = m S_p^2 = X(p) :$$

a positive element — a *pure boost*, by the polar anatomy — with $\Phi(S_p)\hat{p} = p$ and $S_{\hat{p}} = \mathbb{I}$. The whole representation is now three formulas and one telescoping identity.

Theorem 1.8 (The massive representations (m, j)). On $\mathcal{H} = L^2(X_m^+, d\mu) \otimes V_j$ — wave functions $\psi: X_m^+ \rightarrow V_j$, square-integrable against Proposition 1.2 — define

$$(U(a)\psi)(p) = e^{-i\eta(p,a)} \psi(p), \quad (U(A)\psi)(p) = \Pi_j(W(A, p)) \psi(\Lambda^{-1}p), \quad (1.7)$$

where $\Lambda = \Phi(A)$ and

$$W(A, p) := S_p^{-1} A S_{\Lambda^{-1}p} \quad (1.8)$$

is the Wigner rotation. Then $W(A, p) \in SU(2)$, the formulas define a unitary representation $U^{(m,j)}$ of $\mathcal{P}$. Its irreducibility is the imported irreducibility clause of Theorem 1.7. Distinct pairs (m, j) give inequivalent representations, and by Theorem 1.7 these exhaust the massive orbits.

Proof. $W(A, p)$ stabilises $\hat{p}$ — apply Φ : $\Phi(S_p)^{-1} \Lambda \Phi(S_{\Lambda^{-1}p})$ sends $\hat{p} \mapsto \Lambda^{-1}p \mapsto p \mapsto \hat{p}$ — so it lies in the little group, which Proposition 1.5 computed to be exactly $SU(2)$, and Π_j may eat it. The homomorphism property is a telescope:

$$W(A, p) W(A', \Lambda^{-1}p) = S_p^{-1} A (S_{\Lambda^{-1}p} S_{\Lambda^{-1}p}^{-1}) A' S_{(\Lambda\Lambda')^{-1}p} = W(AA', p),$$

whence $U(A)U(A') = U(AA')$ by chasing (1.7); the mixed law $U(A)U(a)U(A)^{-1} = U(\Lambda a)$ is the character identity $\eta(\Lambda^{-1}p, a) = \eta(p, \Lambda a)$; unitarity is Π_j unitary plus the invariance of

$d\mu$. Irreducibility and inequivalence now invoke Theorem 1.7. The relevant commutants are measurable matrix-valued multipliers defined only almost everywhere, so evaluating one at the single point $\hat{p}$ is not a valid elementary proof. The required measurable-field/ergodic reasoning is part of the import and is not silently replaced here. ■

Read the formulas physically. At the reference fibre, rotations act by $\Pi_j(U)$ itself — $W(U, \hat{p}) = U$ — so the index of V_j is literally Lecture 10's spin multiplet, sitting in the rest frame: *spin is the little group's quantum number*, and the classification of Lecture 10, proved as a theorem about $\mathfrak{su}(2)$, has been promoted to the definition of what spin is. For a boost A , the Wigner rotation (1.8) is a product of three boosts that is nevertheless a rotation — precisely the rotation debris of Lecture 14's bracket $[K, K] = -\epsilon L$, now with a job description: it is *how a moving particle's spin frame rotates*, the cocycle that Thomas precession was waiting for. Sheet 14's perpendicular-boost angle is exactly $W(A, p)$ for that configuration, and Notebook 15 watches the telescope identity hold to machine precision.

Remark 1.9 (The massive Casimir values). Two self-adjoint observables commute with the entire representation. The first is $\widehat{P}^2 = \eta^{\mu\nu} \widehat{P}_\mu \widehat{P}_\nu$, which acts in (1.7) as multiplication by $\eta(p, p) = m^2$. The second is built from the self-adjoint Pauli–Lubanski observable

$$\widehat{W}^\mu = \frac{1}{4} \epsilon^{\mu\nu\rho\sigma} \{\widehat{P}_\nu, \widehat{M}_{\rho\sigma}\}, \quad \epsilon^{0123} = 1,$$

on the common invariant domain, where $\widehat{M}_{ab} = -\epsilon_{abc} \widehat{J}_c$. The anticommutator makes symmetry manifest and equals the usual ordered formula after contraction of the Poincaré brackets. One checks $[\widehat{W}^\mu, \widehat{P}_\nu] = 0$ and that $\widehat{W}^2$ commutes with everything (the full index calculation is carried out in Sheet 15, Exercise 2, the Pauli–Lubanski/Casimir capstone). On the rest fibre the momentum factors freeze to $(m, \vec{0})$, the orbital terms drop out, and $\widehat{W} = (0, m\widehat{J})$ with $\widehat{J}$ the rest-frame rotation generators, so

$$\widehat{W}^2 = -m^2 \widehat{J}^2 = -m^2 j(j+1) \mathbb{K}.$$

Thus on the massive positive-energy branch the two eigenvalues recover mass and spin. This conclusion is deliberately orbit-specific: on the finite-helicity massless branch both Casimirs vanish and do not recover helicity.

Two footnotes from earlier lectures come due. The wave functions $\psi(p)$ with their $d^3p/2E_p$ norm are the relativistic normalisation of every field-theory textbook, explained: the $2E_p$ buys invariance (Proposition 1.2). And many particles cost no new group-level construction: the state space of n identical such particles is the symmetric or antisymmetric power of $\mathcal{H}$, and every unitary $U^{(m,j)}(g)$ lifts to the everywhere-defined all-Fock unitary $\Gamma(U^{(m,j)}(g))$. A derived generator is a different statement: it is first $d\Gamma(A)$ on the algebraic finite-particle core, is generally unbounded (already the number operator is), and its skew-adjoint closure generates $\Gamma(e^{tA})$. Thus Lecture 13's second-quantisation machine does not make derived all-Fock generators everywhere defined.

1.5 Massless particles: helicity

The massless story differs in exactly one ingredient: the little group is $\widetilde{E}(2)$, and its representation theory has a fork in it.

Proposition 1.10 (Unitary dual of $\widetilde{E}(2)$; Mackey classification imported). (i) The subgroup $\{\beta\} \cong \mathbb{C}$ acts trivially in finite dimension. Each irreducible is therefore a one-dimensional character of the rotation cover,

$$\chi_h: \text{diag}(e^{-i\theta/2}, e^{i\theta/2}) \mapsto e^{-ih\theta}, \quad h \in \frac{1}{2}\mathbb{Z},$$

single-valued on the double cover, with $\chi_h(-\mathbb{K}) = (-1)^{2h}$ — rays, one last time.

(ii) A nonzero translation character has a circular orbit of radius $\rho > 0$. Its stabiliser in the rotation cover is $\{\alpha : \alpha^2 = 1\} = \{\pm 1\}$, so for every ρ there are two inequivalent infinite-dimensional irreducibles, distinguished by the two characters of this stabiliser (equivalently, by the central sign). These are the continuous-spin representations. With ρ normalised as the Pauli–Lubanski scale, $\widehat{W}^2 = -\rho^2 \mathbb{K}$ in signature $(+ - - -)$.

Proof. The commuting unitaries $U(\beta)$ decompose the space into characters $e^{i\text{Re}(\bar{k}\beta)}$, $k \in \mathbb{C}$, and conjugating by the rotation $\text{diag}(\alpha, \bar{\alpha})$ carries the k -eigenspace to the $\alpha^2 k$ -eigenspace (the semidirect relation of Proposition 1.6, pushed through the representation). The set of occurring labels is therefore invariant under the full circle of rotations; in finite dimensions it is also finite, and the only finite circle-invariant subset of $\mathbb{C}$ is $\{0\}$. With the translations dead, the representation factors through the circle of rotations, whose irreducible representations are the characters listed — the half-integers entering exactly as they did for $\text{SU}(2)$.

For the complete unitary dual, apply the regular semidirect-product theorem separately to $\mathbb{C} \rtimes \text{U}(1)$ with action $\beta \mapsto \alpha^2 \beta$. The dual action sends $k \mapsto \alpha^2 k$. Its orbits are $\{0\}$ and the circles $|k| = \rho > 0$; their stabilisers are respectively $\text{U}(1)$ and $\{\pm 1\}$. The little representations are therefore the characters listed in (i), or one of the two signs in (ii). Folland, Theorem 6.42, supplies induction and irreducibility; Ramond, Ch. 11, §11.1.1, pp. 230–231, records the two cover sectors and the helicity/continuous-spin fork. This little-group classification is not a consequence of merely applying SNAG to the Poincaré translations. ■

The label h is the *helicity*: the eigenvalue of angular momentum projected on the direction of motion, the only rotation a light ray has time for. A massless particle of helicity h is the induced representation over X_0^+ with $\sigma = \chi_h$ — *scalar-valued* wave functions, one component per momentum, with norm

$$\|\psi\|^2 = \int_{\mathbb{R}^3 \setminus \{0\}} |\psi(|\vec{p}|, \vec{p})|^2 \frac{d^3 p}{2|\vec{p}|},$$

using Proposition 1.3, and with a measurable family of standard transformations along the cone (rotation, then boost).

The Casimirs now behave differently. At $\hat{p} = (E, 0, 0, E)$ one has $\widehat{P}_0 = E$ and $\widehat{P}_3 = -E$. In the finite-helicity branch the two null-rotation generators vanish, while $U(r(\theta)) = e^{-i\theta \widehat{J}_3} = e^{-ih\theta}$ gives $\widehat{J}_3 = h$. The defining signs therefore give

$$\widehat{W}^0 = \frac{1}{2} \epsilon^{03jk} \widehat{P}_3 \widehat{M}_{jk} = E \widehat{J}_3, \quad \widehat{W}^3 = \frac{1}{2} \epsilon^{30jk} \widehat{P}_0 \widehat{M}_{jk} = E \widehat{J}_3,$$

and $\widehat{W}^1 = \widehat{W}^2 = 0$. Hence

$$\widehat{W}^\mu = h \widehat{P}^\mu \quad \text{on every finite-helicity massless representation,} \quad \widehat{P}^2 = \widehat{W}^2 = 0. \quad (1.9)$$

The sign is fixed, not remembered: both displayed nonzero components are $+Eh$, and covariance carries the relation along X_0^+ . Thus the two Casimir eigenvalues do not distinguish the different values of h .

Nature's choice at the fork is empirical and unanimous: every massless particle observed is of type (i). The continuous-spin option — infinitely many polarisation states at each momentum — is a chair at Wigner's table that has never been filled.

Helicity's physics is its behaviour under parity. Spatial inversion reverses $\vec{p}$ but not the axial $\vec{J}$, so it flips $\vec{J} \cdot \hat{p}$: helicity h becomes $-h$ (at group level, conjugation by parity carries the little group of $\hat{p}$ to that of the reversed ray, inverting the rotation character). A parity-symmetric theory must therefore contain $\pm h$ together, and *electromagnetism is the example*: the photon is the pair $h = \pm 1$, and so has exactly *two* polarisation states — where a *massive* spin-one particle has $2j + 1 = 3$. The missing longitudinal photon is a little-group theorem: the third state is a luxury only mass can buy. (The graviton's pair is $h = \pm 2$; the weak interaction, which violates parity, happily employs unpaired helicities.) One last ledger entry: the photon *state* has one component per momentum and helicity, while the photon *field* A_μ has four — the mismatch between $(\frac{1}{2}, \frac{1}{2})$ and $\chi_{\pm 1}$ is real, and the apparatus that manages it is gauge invariance: the final, and deepest, item in the fields-versus-states accounts, payable in a field theory course.

1.6 Retrospect

The positive-energy catalogue is now separated into its branches. An elementary relativistic system is an orbit and a little-group representation: massive particles (m, j) , with spin defined by the rest frame's $SU(2)$; massless particles $(0, h)$, with helicity defined by the ray's $\tilde{E}(2)$ rotation character; the two continuous-spin families at each $\rho > 0$, distinguished by the cover's central sign; and the vacuum, alone on the trivial orbit. In the empirical table below two chairs stand empty: the continuous-spin representations, and the spacelike orbits — whose little group, the stabiliser of $X(\hat{p}) = \mu\sigma_3$ in $SL(2, \mathbb{C})$, is the noncompact group of σ_3 -unitaries $SU(1, 1) \cong SL(2, \mathbb{R})$, the double cover of the geometric little group $SO^+(1, 2)$, so that nontrivial unitary little-group representations would give tachyons infinitely many spin states on top of their causal misbehaviour. The trivial unitary representation is the one-dimensional exception. Nature, as far as anyone has looked, declines both.

The dictionary to the particles of nature is then read off a single table:

Wigner label	little group	physical example
$(m, 0)$	$SU(2)$	the Higgs boson
$(m, \frac{1}{2})$	$SU(2)$	electron, quarks, massive neutrinos
$(m, 1)$	$SU(2)$	the $W^\pm$ and Z bosons
$(0, \pm 1)$	$\tilde{E}(2)$	the photon, the gluons
$(0, \pm 2)$	$\tilde{E}(2)$	the (conjectured) graviton
$(0, 0)$	$SL(2, \mathbb{C})$ (trivial irrep)	the vacuum

The one historical correction sits in the second row: neutrinos were long entered as massless $(0, \pm \frac{1}{2})$, but flavour oscillations prove $m_\nu \neq 0$, moving them onto the massive ladder — the single empirical edit made to Wigner's 1939 table.

What remains between this classification and physics is one construction: a quantum *field* is a spacetime-indexed family of Fock-space operators intertwining a finite-dimensional (j_1, j_2) — Lecture 14's fields — with the infinite-dimensional unitary representations built today — Lecture 15's states. The Dirac equation was the prototype of such an intertwining condition; building them systematically, with locality and causality enforced, is quantum field theory, the natural next course. Its multi-particle arena, Fock space with Γ and $d\Gamma$, left Lecture 13 ready for use.

The course's SU(3) boundary is deliberate. Duals and conjugate representations, invariant tensors, SU(2) coupling, and highest-weight staircases have prepared the vocabulary and methods. Applying them to the quark $\mathbf{3}$ and antiquark $\bar{\mathbf{3}}$, the eightfold way, roots and weights, and strong-interaction invariant tensors is further study, not material taught in these lectures. Jones, Ch. 8, §§8.3–8.4, pp. 149–164 is the exact next destination. The other horizons are visible from here: symmetry made *local* (gauge theory), symmetry *broken* by a ground state, harmonic analysis and Mackey theory in earnest, and the strange cases where a classical symmetry fails to survive quantisation.

We opened Part I by counting the symmetries of a triangle; we close Part II by defining what a particle is. It was the same calculation throughout — decompose, then intertwine — performed with heavier measures and steadier nerves, and its moral has not changed since the third week: a symmetry is never just a group; it is a group *represented*, and the representation is where the physics lives.

Remark 1.11 (Computer algebra). Notebook 15 draws the orbit catalogue (sheets, cone, hyperboloid); computes standard boosts S_p by matrix square roots and verifies $\Phi(S_p)\hat{p} = p$; checks the Wigner cocycle (1.8) and its telescoping identity on random group elements to machine precision; tracks the helicity phase $e^{-ih\theta}$ around a full 4π loop of rotations; and confirms the invariance of $d^3p/2E_p$ by Monte Carlo against random boosts.

Summary.

- *Poincaré group.* The semidirect product $\mathbb{R}^{1,3} \rtimes \text{SO}^+(1, 3)$ of translations and Lorentz transformations (eq. (1.1)).
- *Orbits.* The Lorentz orbits in momentum space are the mass shells, the light cone, the spacelike sheet, and the origin (Proposition 1.1); the massive-shell and punctured-cone measures are derived separately in Propositions 1.2 and 1.3.
- *Little groups.* The stabiliser of a momentum is SU(2) for a massive particle (Proposition 1.5) and $\tilde{\text{E}}(2)$ for a massless one (Proposition 1.6).
- *Wigner classification.* An elementary particle is an orbit together with an irreducible representation of its little group (Mackey, Theorem 1.7): massive (m, j) with spin (Theorem 1.8); massless finite-helicity characters $(0, h)$; and two continuous-spin sectors for each $\rho > 0$.
- *Casimirs.* On the massive branch, $\hat{P}^2 = m^2$ and $\hat{W}^2 = -m^2 j(j+1)$ recover mass and spin. On the finite-helicity massless branch $\hat{W}^\mu = h\hat{P}^\mu$, so both Casimirs vanish and the rotation character, not the Casimirs, records h .

Tutorial. Exercise Sheet 15.

References

Folland, 2nd ed., §6.6, Theorem 6.42, pp. 199–201 (regular Mackey theorem); Ramond, Ch. 11, §11.1.1, pp. 224–231 (Casimirs and covered massless sectors); Tung, Ch. 10, §10.4, pp. 191–200 (Wigner classification and Pauli–Lubanski relations); Jones, 2nd ed., Ch. 10, §10.4, pp. 208–216 and Appendix E, pp. 279–280 (little groups and on-shell measure). SNAG: Berndt, Theorem 5.2, p. 71; Barut–Rączka, Ch. 5. SU(3) further reading: Jones, Ch. 8, §§8.3–8.4, pp. 149–164.

Bibliography

The per-lecture *References* sections cite the works below by author and short title; full details are collected here. Chapter and section numbers in the lecture references follow these editions.

- M. A. Armstrong, *Groups and Symmetry*. Springer (UTM), 1988.
- V. Bargmann, On unitary ray representations of continuous groups. *Annals of Mathematics* **59** (1954), 1–46.
- A. O. Barut and R. Rączka, *Theory of Group Representations and Applications*, 2nd ed. World Scientific, 1986.
- R. Berndt, *Representations of Linear Groups: An Introduction Based on Examples from Physics and Number Theory*. Vieweg, 2007. DOI: 10.1007/978-3-8348-9401-4. Theorem 5.2, p. 71 gives the SNAG theorem used for the translation subgroup.
- D. M. Bishop, *Group Theory and Chemistry*. Dover, 1993 (orig. Oxford University Press, 1973).
- O. Bratteli and D. W. Robinson, *Operator Algebras and Quantum Statistical Mechanics 2: Equilibrium States, Models in Quantum Statistical Mechanics*, 2nd ed. (1997), 2nd printing. Springer, 2002. Section 5.2.1, pp. 7–8 gives Fock-space second quantisation, its finite-particle analytic core and closure, and the lifted unitary group.
- G. Costa and G. Fogli, *Symmetries and Group Theory in Particle Physics*. Springer (Lecture Notes in Physics 823), 2012.
- J. B. Conway, *A Course in Functional Analysis*, 1st ed. Springer (GTM 96), 1985. Chapter IX, Theorems 6.4 and 6.6 give the strong/weak bicommutant closure and the multiplication maximal-abelian theorem for sigma-finite measure spaces.
- J. P. Elliott and P. G. Dawber, *Symmetry in Physics*, 2 vols. Macmillan, 1979.
- K. Erdmann and M. J. Wildon, *Introduction to Lie Algebras*. Springer (SUMS), 2006.
- W. Fulton and J. Harris, *Representation Theory: A First Course*. Springer (GTM 129), 1991.
- G. B. Folland, *Harmonic Analysis in Phase Space*. Princeton University Press (Annals of Mathematics Studies 122), 1989. Chapter 1, Sections 2–5, including Theorem 1.50, treats the Heisenberg group, Schrödinger/Weyl representation, and Stone–von Neumann theorem.
- G. B. Folland, *A Course in Abstract Harmonic Analysis*, 2nd ed. Chapman & Hall/CRC, 2016. Section 6.6, Theorem 6.42, pp. 199–201 gives the regular semidirect-product Mackey theorem, including induced irreducibility, exhaustivity, and equivalence.
- R. Gilmore, *Lie Groups, Lie Algebras, and Some of Their Applications*. Wiley, 1974 (repr. Dover, 2005).
- B. C. Hall, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, 2nd ed. Springer (GTM 222), 2015.
- N. Jeevanjee, *An Introduction to Tensors and Group Theory for Physicists*, 2nd ed. Birkhäuser, 2015.
- H. F. Jones, *Groups, Representations and Physics*, 2nd ed. Institute of Physics Publishing,

1998. Chapter 8, Sections 8.3–8.4, pp. 149–164 is the course’s explicit $SU(3)$ further-reading route; Chapter 10, Section 10.4, pp. 208–216 treats the Poincaré little groups; Appendix E, pp. 279–280 derives the covariant on-shell measure from $\theta(p^0)\delta(p^2 - m^2)d^4p$.
- Z.-Q. Ma, *Group Theory for Physicists*. World Scientific, 2007.
 - A. M. L. Messiah and O. W. Greenberg, Symmetrization Postulate and Its Experimental Foundation. *Physical Review* **136** (1964), B248–B267. DOI: <https://doi.org/10.1103/PhysRev.136.B248>.
 - H. Osborn, *Group Theory*. University of Cambridge, Mathematical Tripos Part III lecture notes (available from the DAMTP lecture-notes pages).
 - M. Reed and B. Simon, *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*, rev. ed. Academic Press, 1980. Chapter VIII, Theorems VIII.5, VIII.8, VIII.10, and VIII.13 and Section VIII.5, Example 2 cover spectral calculus, Stone’s theorem, invariant-domain generators, strong commutation, and the bounded-CCR obstruction.
 - P. Ramond, *Group Theory: A Physicist’s Survey*. Cambridge University Press, 2010. Chapter 11, Section 11.1.1, pp. 224–231 treats the Pauli–Lubanski Casimirs and the massless little group; pp. 230–231 distinguish finite helicity from continuous spin and record the two integer/half-odd-integer cover sectors.
 - H. Rauch, A. Zeilinger, G. Badurek, A. Wilfing, W. Bauspiess, and U. Bonse, Verification of Coherent Spinor Rotation of Fermions. *Physics Letters A* **54** (1975), 425–427. DOI: 10.1016/0375-9601(75)90798-7.
 - J. Schwichtenberg, *Physics from Symmetry*, 2nd ed. Springer, 2018.
 - J.-P. Serre, *Linear Representations of Finite Groups*. Springer (GTM 42), 1977.
 - J. Stillwell, *Naive Lie Theory*. Springer (UTM), 2008.
 - E. M. Stein and R. Shakarchi, *Fourier Analysis: An Introduction*. Princeton University Press (Princeton Lectures in Analysis I), 2003. Chapter 3, Section 1, Theorem 1.1, pp. 69–79 gives the mean-square completeness result used in Lecture 12.
 - M. Tinkham, *Group Theory and Quantum Mechanics*. McGraw–Hill, 1964 (repr. Dover, 2003).
 - W.-K. Tung, *Group Theory in Physics*. World Scientific, 1985. Chapter 10, Section 10.4, pp. 191–200 treats induced Poincaré representations, Pauli–Lubanski brackets, and massive and massless little groups.
 - S. Weinberg, *The Quantum Theory of Fields, Vol. I: Foundations*. Cambridge University Press, 1995. Section 5.6 treats causal fields and the relativistic spin–statistics connection, a theorem not derived from a rotation phase in this course.
 - S. A. Werner, R. Colella, A. W. Overhauser, and C. F. Eagen, Observation of the Phase Shift of a Neutron Due to Precession in a Magnetic Field. *Physical Review Letters* **35** (1975), 1053–1055. DOI: 10.1103/PhysRevLett.35.1053.
 - P. Woit, *Quantum Theory, Groups and Representations: An Introduction*. Springer, 2017.